EDA Findings

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Introduction

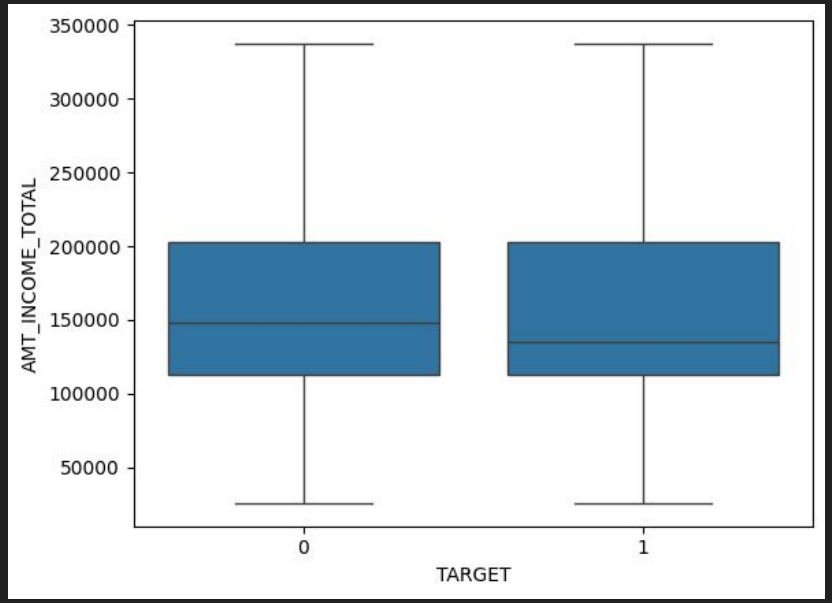
Our primary approach to the EDA of the loan datasets was graphing many different variables against mean values for ‘TARGET’ in order to determine the most predictive variables for default rate – or those which showed the greatest disparity in ‘TARGET’ values across the categories of the variable.

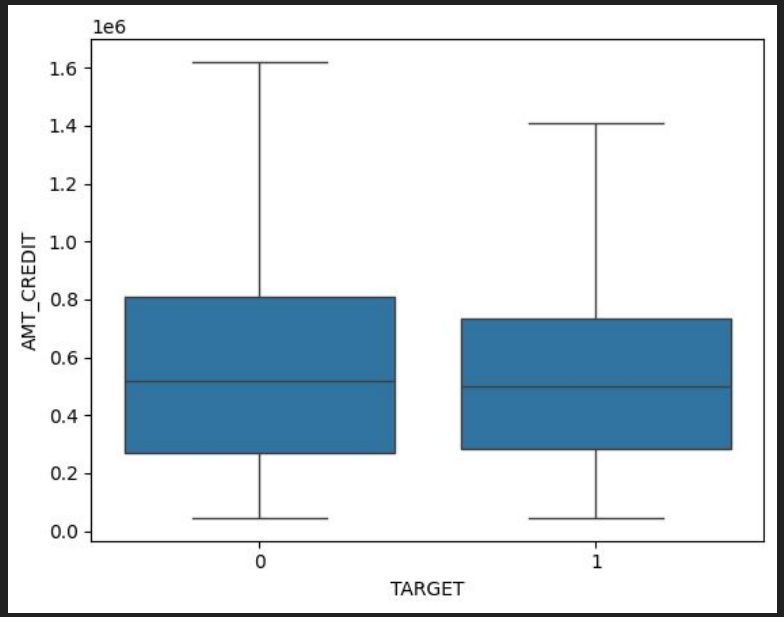
We also merged the two datasets by ID, allowing us to compare current loan data against historical loan performance and application history, which too proved to be important in determining the creditworthiness of the individuals in the current dataset.

General Conclusions

- Ironically, common factors such as income, credit request history, and credit amounts provided little visibility into the probability of default
  - Factors considered as an extension of income (housing, job type, city tier, and education) were more enlightening than pure income
- Credit scores remain a strong predictor of credit-worthiness.
- Candidates who had a lower credit score or shorter employment history tended to default at a greater rate.
- From previous data, customers who were previously refused credit, as well as those refused or cancelled specifically due to a high interest rate, were more likely to have difficulties paying the loan in question.

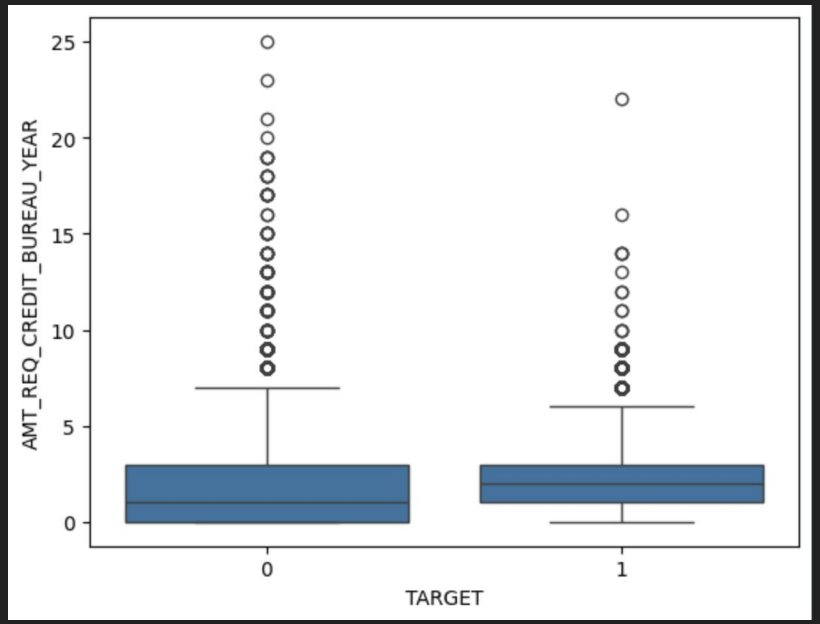
Income and Credit Amount





While defaulters tend to have lower median incomes, the income alone provides little visibility into default rate.
Loans people tended to default on were of considerably smaller amounts, but credit amounts did not reveal a meaningful conclusion.

Credit Request History (Trailing Twelve Months)



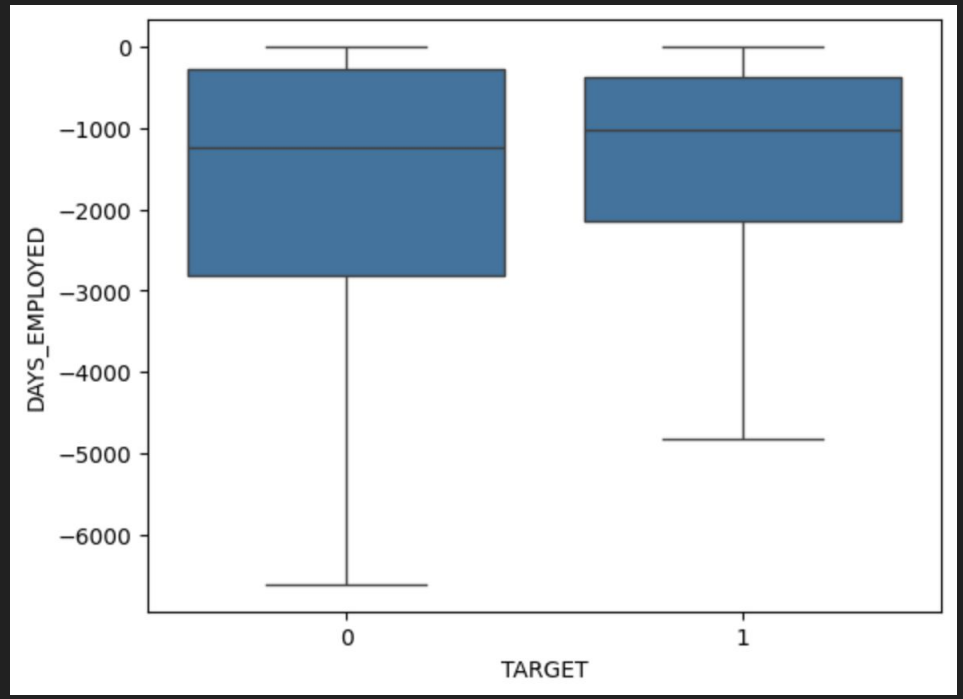
While defaulters had a slightly higher mean number of requests for new credit over the previous year, and number of high outliers and higher Q3 for those who successfully paid off their loans draws into question the usefulness of this metric.

In sum, the number of requests for new credit, which factors into one's credit score, does not really seem to impact the creditworthiness of an individual all that much.

Employment History

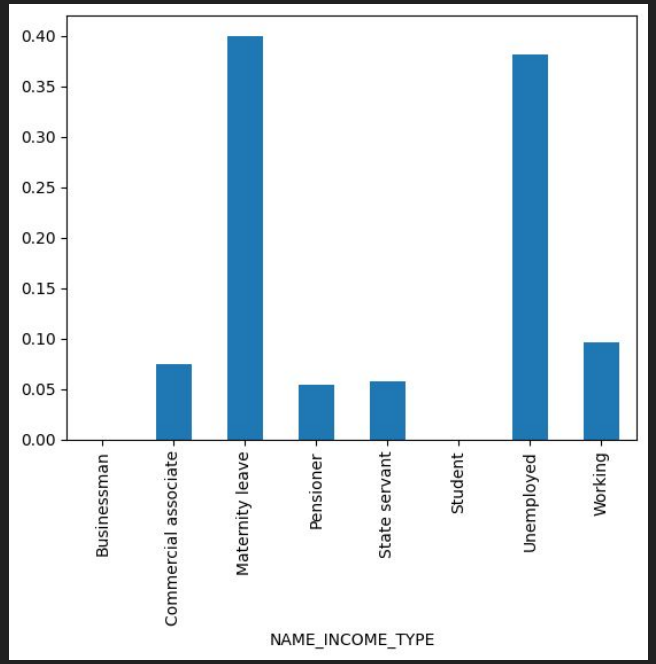
Examining these boxplots, those who performed well on the loan in question have typically been employed far longer than those who had difficulties paying.

As a variable, days of employment is also tied to other factors such as age, income, job stability, etc. that would likely be more determinate of creditworthiness. However, it remains an important factor for lenders to consider given the stark differences in the graph.

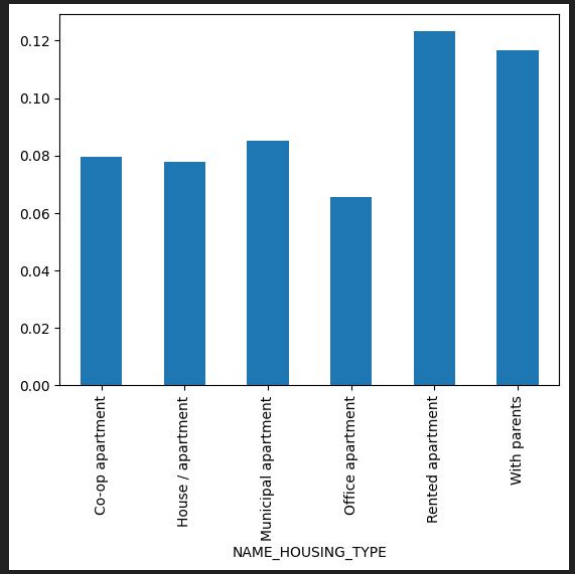


Income-Adjacent Indicators

Naturally, unemployment (including maternity leave) was correlated with a higher default level (no income stream). Additionally, housing status (with apartments indicating lower average income) revealed, though the results were quite dispersed, there was considerable volume living in apartments or with parents.



INCOME TYPE VS TARGET



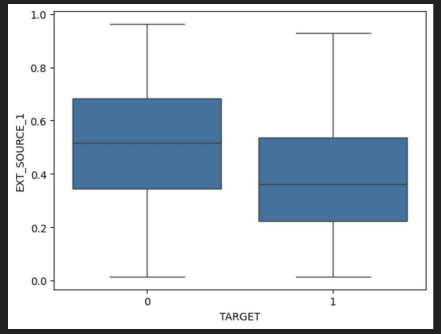
HOUSING VS TARGET

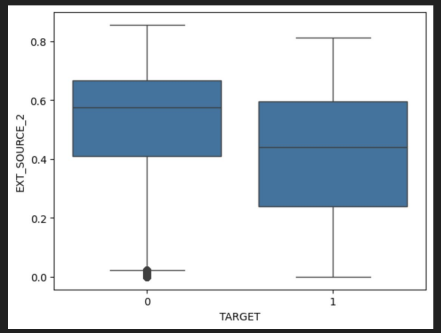
External Sources/Credit Scores

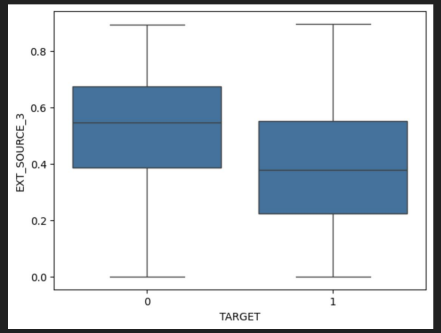
The predictive value of credit scores showed similar results across the three credit bureaus.

Universally, those who faced payment difficulties on their credit typically had significantly lower credit scores than those who were able to pay of their loans.

This analysis only proves the predictive value of credit scores for the financial stability of an individual, and how important they are for the process of getting credit.





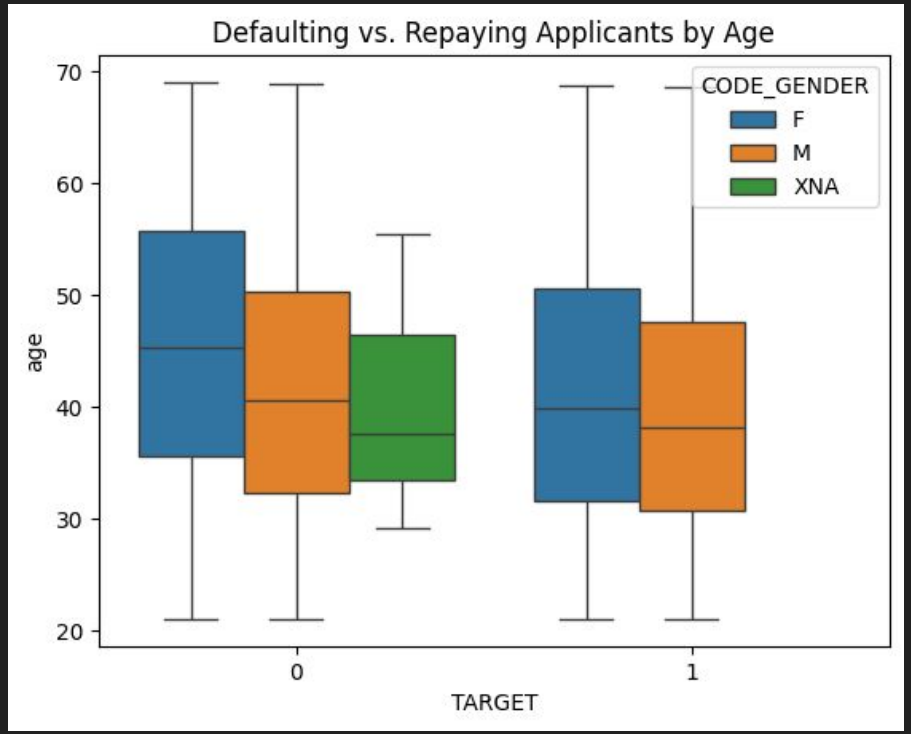


Age Distribution and Gender Profiles across Loan Outcomes

Age as a Primary Risk Factor: There is a visible downward shift in the age distribution for the Default (Target 1) group compared to the Repayment (Target 0) group. This indicates that younger applicants are more likely to struggle with loan repayment.

Male applicants show a lower median age in both categories, suggesting that the "risk zone" for men starts earlier than it does for women.

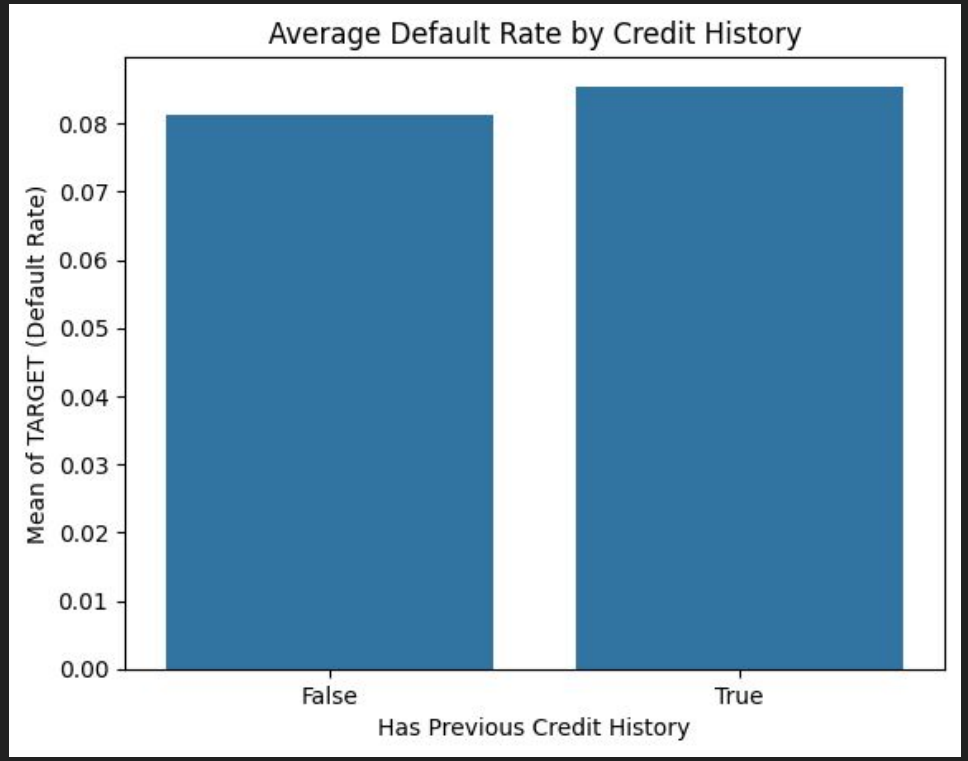
Female applicants who default (Target 1) still have a higher median age than males who repay (Target 0), showing that gender is a strong secondary factor alongside age.



Default Rates by Previous Application History

Surprisingly, applicants with a known history in the system actually have a higher default rate (8.54%) than those without any history (8.12%).

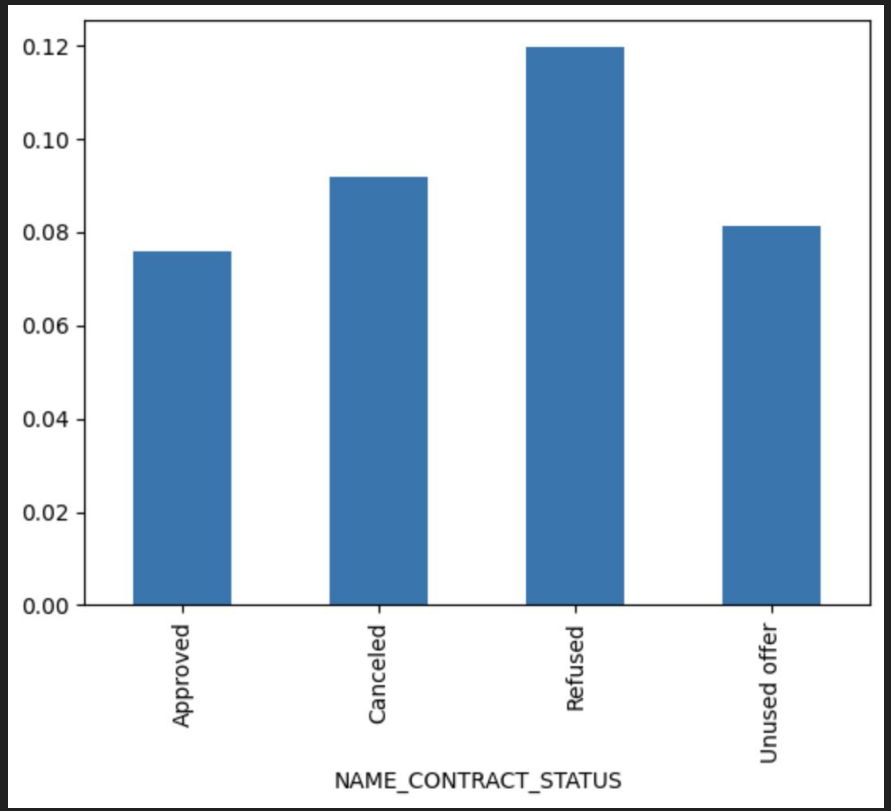
While the difference is small (about 0.42%), it suggests that simply *having* a previous application doesn't automatically make someone a safer bet.



Previous Loan Data

Those who were approved on a previous loan were less likely to miss payments than those who cancelled their offer, who were substantially less likely than those previously refused for a credit offer.

Logically, if someone has been refused a loan before, it was likely due to their prospective inability to pay off said loan. Therefore, the credit history is a strong predictor of future credit performance.



Conclusion

What should banks look at when deciding whether or not to lend?

- **Previous Loan Outcomes**
- **External Credit Scores**
- **Applicant Age**
- **Previous Application Behavior**
- **Credit History Depth**

What do banks typically look at that may not be all that useful?

- **Binary "Has History" Flags**
- **Standalone Income Levels**
- **Specific Credit Amounts**
- **Simple Employment/Income Metrics**